Exercise – 11: Work on Different Types of Vulnerabilities

Aim

To study and analyze **different types of vulnerabilities in computer systems and networks** and understand how they can be identified and mitigated.

Objectives

- To understand the concept of **system vulnerabilities**.
- To identify different types of vulnerabilities in applications, networks, and systems.
- To learn how vulnerability scanning tools help detect security weaknesses.

Software Requirements

- Kali Linux / Ubuntu / Windows
- Vulnerability scanning tools:
    - Nmap
    - OpenVAS / Greenbone Vulnerability Manager
    - Nikto (Web vulnerability scanner)

Hardware Requirements

- Computer system
- Network connection

Theory

Vulnerability

A vulnerability is a **weakness or flaw in a system, software, or network** that can be exploited by attackers to gain unauthorized access or perform malicious actions.

Attackers use vulnerabilities to:

- Access sensitive information
- Modify data
- Install malware
- Disrupt services

Vulnerabilities may exist due to:

- Programming errors
- Poor security configuration
- Outdated software
- Weak passwords

Types of Vulnerabilities

1. Software Vulnerabilities

These occur due to **bugs or errors in software programs**.

Examples:

- Buffer overflow
- Improper input validation

2. Network Vulnerabilities
Weaknesses in network infrastructure.
Examples:

- Open ports
- Weak network protocols

3. Web Application Vulnerabilities
Common web vulnerabilities include:

- SQL Injection
- Cross Site Scripting (XSS)
- Broken authentication

4. Configuration Vulnerabilities
Misconfigured systems can expose sensitive data.
Examples:

- Default passwords
- Unnecessary services enabled

Tool 1: Nmap (Network Scanner)
Purpose
Nmap is used to **identify open ports, services, and potential vulnerabilities in a network system**.

Steps to Use Nmap
Step 1: Open Terminal
Open terminal in Kali Linux.

Step 2: Scan a Target System
nmap <target-ip>
Example:
nmap 192.168.1.1
This command scans open ports on the target system.

Step 3: Service Version Detection
nmap -sV <target-ip>
This command detects services running on open ports.

Step 4: Vulnerability Scan
nmap --script vuln <target-ip>
This command checks for known vulnerabilities.

Tool 2: Nikto (Web Vulnerability Scanner)
Nikto scans web servers for security issues.

Steps
Step 1: Open terminal
Step 2: Run Nikto scan
nikto -h http://targetwebsite.com
Nikto will detect:

- Outdated server software
- Dangerous files
- Server misconfigurations

Tool 3: OpenVAS (Vulnerability Scanner)
OpenVAS is a vulnerability management tool used to detect **security weaknesses in systems and networks**.

Steps to Perform Scan

1. Start OpenVAS service.
2. Open browser and access the OpenVAS web interface.
3. Create a new **scan target**.
4. Start vulnerability scanning.
5. Analyze the scan report.

Observation

Tool Used	Vulnerability Identified
Nmap	Open ports and running services
Nikto	Web server vulnerabilities
OpenVAS	System security weaknesses

Applications

- Security auditing
- Penetration testing
- Risk assessment
- Network security monitoring

Precautions

- Perform scans only on **authorized systems**.
- Avoid scanning production systems without permission.
- Follow ethical hacking guidelines.

Result

Thus, different types of vulnerabilities in systems and networks were successfully identified and analyzed using vulnerability scanning tools.

Viva Questions

1. What is a vulnerability?
2. What is vulnerability scanning?
3. What is Nmap used for?
4. What is Nikto?
5. What is the purpose of OpenVAS?
6. What are common web vulnerabilities?